



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.AT.5

Explore Numerical Expressions with Exponents

Lesson 6-3 • Teacher Copy

Name: _____

Date: _____

Class: _____

SOUND STUDIO MISSION

The Doubling Mixboard

You are the lead sound engineer at Amplitude Studios. Every time you flip a switch on the new mixboard, the volume doubles — that is the power of exponents. The headline act goes on in one hour, and you must master powers before the speakers can be safely dialed in.



TODAY'S OBJECTIVES

Content Objective: I can write and evaluate numbers in exponent form using a base and a power.

Language Objective: I can explain my work using the words exponent, base, power, and evaluate.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Exponent

Base

Power

Evaluate

Repeated multiplication

Expanded form

Squared

Cubed



Tap any word to see its meaning and a picture.

1

The small raised number that shows how many times the base is multiplied is the

2

The number being multiplied repeatedly in a power, like the 3 in 3^4 , is the

3

A number written with a base and an exponent, like 3^4 , is a

4

To find the value of an expression, like $2^3 = 8$, is to

it.

5

$5 \times 5 \times 5 \times 5$ is an example of

6 Writing 2^3 as $2 \times 2 \times 2$ shows its .

7 A number used as a factor two times is .

8 A number used as a factor three times is .

Write About the Math

How large is the file after 5 layers? Why does the file size grow so quickly?

¿De qué tamaño es el archivo después de 5 capas? ¿Por qué crece tan rápido el tamaño del archivo?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Exponent**
Exponente

☐ **Base**
Base

☐ **Power**
Potencia

☐ **Evaluate**
Evaluar

☐ **Repeated multiplication**
Multiplicación repetida

Model: *The exponent counts how many times we multiply, not what we add.* — Students connect each switch flip to one more factor of 2 and explain that $2^3 = 2 \times 2 \times 2 = 8$, distinguishing repeated multiplication from repeated addition (which would give 6).

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **After 5 layers the file is ____ MB because $4 \times 2^5 = 4 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$. The file grows quickly because each layer ____ the size.**
- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **My evidence is ____, so ____.**
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Exponent
2. **Notes 2:** Base
3. **Notes 3:** Power
4. **Notes 4:** Evaluate
5. **Notes 5:** Repeated multiplication
6. **Notes 6:** Expanded form
7. **Notes 7:** Squared
8. **Notes 8:** Cubed
9. **Watch for this mistake:** *A common mistake in Powers and Exponents is multiplying the base by the exponent instead of using the base as a repeated factor — for example, thinking $2^3 = 2 \times 3 = 6$ instead of $2^3 = 2 \times 2 \times 2 = 8$. Before you submit, rewrite the power as repeated multiplication first, then multiply step by step.*
10. **Try It 1:** B. 7^3 — *The base 7 is used as a factor 3 times, so the power is 7^3 . $7^3 = 7 \times 7 \times 7 = 343$.*
11. **Try It 2:** D. 64 — $2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$. *The exponent 6 counts how many times 2 is a factor.*